

Use the pre-worksheet to collect information that may help with filling out the scorecards.

[DOI: 10.5281/zenodo.15122045](https://doi.org/10.5281/zenodo.15122045)

[Repository Crisis Scorecards Instructions](#) Date:

Descriptor	Question	Answer
<i>Total metadata/data volume</i>	<i>What is the total volume of metadata and data?</i>	
<i>Bandwidth availability</i>	<i>How much bandwidth is available for upload and download?</i>	
<i>Total irreplaceable datasets</i>	<i>For how many datasets is this repository the primary (only?) archive? What datasets need to be prioritized because they can't be obtained elsewhere?</i>	
<i>Metadata schemas</i>	<i>How many and which metadata schemas can be outputted?</i>	
<i>Dataset Packaging</i>	<i>Is it possible to download an archive copy of all datasets with all metadata attached?</i>	
<i>Data types</i>	<i>Can datasets be regenerated in the case of loss? What data types are held, ex. observations, model analysis, others.</i>	
<i>Dynamic (near real-time) or static datasets</i>	<i>Do you host dynamic datasets, and if so how often do they change?</i>	

	What sensitivities are there around the data? (e.g., restricted data, indigenous data, etc)
Sensitive data considerations	
Are metadata and data stored separately?	What are the strategies for backups and how are data and metadata harmonized?
	What are critical cyberinfrastructure services such as databases and hardware maintenance, and are they administered on-premise or a commercial service managed elsewhere/software as a service (SAAS), such as in the cloud?
Hosted Software Location	
	Are there links to public plans detailing disaster recovery, data migration, and repository shutdown?
Public Plans	
	What, if any, are your service level agreements to keep and/or restore service during crisis and non-crisis periods?
Service Level Agreements	

Repository Name:
Repository Parent
Organization:
Project:

Resiliency Considerations

The total volume of metadata and data can influence the ease and speed with which data and metadata can be replicated, and can be important to discuss when collaborating with other repositories for arranging backups. Having this information readily available can be useful in making these plans.

Knowing the available bandwidth can help in planning a schedule of data backups or planning for replication resources in the event of a crisis.

Understanding the proportion of resources that are unique is important in evaluating the risk of loss of information. This can help in prioritizing efforts and time.

Ex: Frictionless Data Packa

In the case of a repository holding multiple projects or a wide variety of data types, it may be helpful to fill out multiple paragraphs or multiple scorecards for each project.

When setting up an alternate data host, consideration should be given to whether there are proprietary or sovereignty considerations that need to be accounted for (e.g. via restricted access log-in). In addition, any personally identifiable information (PII) should continue to be managed according to requirements/regulations and best data management & governance practices.

[List webpages or DOIs to pl](#)

The Repository Resilience Scorecard is used to help digital data repository staff assess cri
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Repository Crisis Scorecards Instructions

Repository Descriptors			Resiliency Metric	
			No resiliency	
Big Picture Question	Descriptor	Descriptor definition	Class 1	Class 2

Section Theme: Single Repository Dataset Replication

This section is filled out from the viewpoint of the organization looking at its dataset

How many dataset copies does the repository have?	Total number of logical copies	Number of discrete logical copies ma	One copy on one physical device	Multiple copies on one physical device
	Total number of storage medium types	Digital storage medium physical devices includes: hard disk drives, solid state drives, tape drives, Blu- ray discs, cloud storage	One type of storage medium	Two types of storage medium
	Total number of offline copies	Dataset copies that are held offline in an environment controlled, secure location to prevent bit rot or tampering.	Zero offline copies	One offline copy
	Total number of national repository facilities storing copies	Facilities are managed by the repository and are geographically separated and grid separated, ex. Los Angeles, USA and Richmond, USA.	One data facility	Two data facilities, geographically separated
	Total number of extranational repository facilities storing copies	Facilities are managed by the repository and are geographically separated and grid separated, ex. Los Angeles, USA and Tokyo, Japan. The facilities could be self- owned or a commercial cloud service.	Zero international data facilities	One international data facility

Section Theme: Repository Infrastructure

Repository Staff and Training	Are there redundant staff and requisite training for the repository? Can staff cover for each other in case of unavailability?	Each staff member is specialized and unable to do each other's role, or there is only one staff member.	There is one staff member in each role and staff in each role can cover for staff in other roles.
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Can the physical operations of the repository run smoothly when in a crisis?	Repository Supported Virtual Data Transfer	Do bandwidth limitations impede data transfer options for critical datasets from the repository?	Data volume is too large to effectively transfer all data within one month.	Data volume is too large to effectively transfer all data within one week.
	Repository Supported Physical Data Transfer	What is the capability to send datasets physically in an emergency, ex. hard drives in a van?	There are no plans or training to physically move datasets, and/or it would take more than 48 hours.	There is capability to break down and physically move datasets in two days notice.
	Physical Plant Infrastructure Capability	How stable are power, HVAC, plumbing, building maintenance? (Should this question be split further?)	There are no redundant systems at any level.	Some systems have redundancy, others don't.

Section Theme: Routine Operations

Are processes and documentation in place to ensure resiliency on a routine basis?	Backup Schedule	How often are backups performed of the repository?	No regular updates or backups, or no device to backup to.	Backups are manual, irregular, and on the same device, and/or backups are not tested regularly.
	Service Level Agreements	Are there Service Level Agreements (SLAs) in place to communicate to users when they can expect services to be resumed in case of problems?	No SLAs are defined.	SLAs are defined for some operations.
	Continued Training/Professional Development	Do staff have regular training opportunities?	No required training for repository work or time allocated for staff improvement.	Required training for staff but no time/funds for staff improvement via optional conferences, trainings, or certifications.

Section Theme: Software

Is the software currently being used to manage datasets integral to the accessibility of the holdings? Could this software be replaced with other software easily when considering potential costs to use and maintain the software, and use and maintain the holdings? Is

Software as Requirement for Interpreting Datasets	Is the software a critical piece needed to understand and interact with the data and metadata? Ex. NetCDF library and tools	The software is integral to using the data and metadata efficiently, and datasets would not be accessible without the software.	There are multiple pieces of software that can be used to interact with the data and metadata that are interchangeable.
Software Packaging and Backup Format	How are the systems software delivered by the developers and backed up by the repository? Is it possible for the public to get the software?	The software is not compiled by the developer to be delivered as a binary executable, and may be multiple files spread across directories.	
Software Availability and Licensing	Are the systems software open source and unencumbered by licenses? If proprietary, how easy is it to acquire a license?	The software is proprietary and it is hard or impossible to get a license to use it.	The software is open source but has been abandoned/no updates to it in a while, OR the software is proprietary but still being supported by a company.
Ease of Software Deployment	Is the software well documented and easy to install? Are there images or virtual machines to ease deployment? Are there sane initial configurations to use the software?	There is very little code and supporting documentation. Source code is difficult to understand and manage or not available.	Source code is well documented and easy to install by a technical user. Configuration may take some time. There is very little support for non-technical users to deploy the software.
Maintainability of Software	How easy is it to maintain and extend the code?	The software is written in a language(s) that are not as popular and/or leverage unmaintained or expensive libraries. Code discipline is weak and the codebase would be greatly improved by a rewrite.	The software is in a common language and has decent code discipline, documentation, and testing. The software may rely on some libraries with weak support, or need some code changes to be supported by more robust communities.
Ease of Use of Software	How easy is it for an experienced non-programming user to work with the software to curate and access datasets?	Curators and users require a long training period to use the software. Documentation is lacking or non-existent. Programmatic access is lacking or non-existent.	Curators or users might have trouble using the software. Documentation exists but could be filled out. Some defaults are provided when performing common actions.

the software publicly accessible outside of the repository?

Section Theme: Cybersecurity and Access Control

Logging and Change Monitoring	How are changes tracked or logged? Can changes be rolled-back? What processes exist to ensure unauthorized changes are noted, alerted, and corrected?	There is no automated logging activated at the OS or repository systems level, and manual logging of actions is ad hoc.	There is automated logging at the OS level. There may be automated logging at the repository systems level, but monitoring of logs are ad hoc and infrequent. Logs may not be comprehensive, understandable, or easily accessible.
Access Control and Deletion Privileges	Are permissions appropriately scoped for each account? Are policies in place to limit accidental or hostile access to critical systems? Are editing and deletion privileges restricted to authorized users and clearly documented?	Staff permissions are not differentiated. There are no clear policies to follow for deletion of deposits, and no logs or records of deletions exist.	There are some accounts with higher permissions allowed to delete data. Some policies may be defined for deletion of data, but no additional safeguards.
Cybersecurity Staffing	Are there dedicated roles for cybersecurity, and how many are there in the repository and larger organization? Do cybersecurity personnel have appropriate qualifications?	There are no cybersecurity professionals on staff or available through a larger organization.	There are cybersecurity professionals available through a larger organization.

Are best practices in place to ensure cybersecurity?

Section Theme: Sustainability

What agreements and plans are in place to preserve the mission of the repository in the face of crisis or after cessation of activities?	Data or Systems Modernization or Migration Plans	Does the repository have plans for the modernization of non versionless (ex. RAW image format, text) datasets and systems to new formats, tools, audiences?	Datasets/data systems are frozen, no data modernization capacity.	Dataset/data systems modernization tools and/or tutorials are available, but new versions of datasets are not hosted.
	Repository Shutdown Plans	Does the repository have a clear plan for systematic shutdown of systems that preserves the data?	No shutdown plan exists.	There are plans for a systematic shutdown of systems but not transfer of datasets.
	Continuity Agreement (Intranational)	Does the repository have a documented agreement for relocation or transition of holdings to another intranational repository in the event that the repository can no longer fulfill its mandate due to funding or shifted priorities?	No documented plan or agreement exists.	There is a formal agreement but lacking detailed plans.
	Continuity Agreement (Extranational)	Does the repository have a documented agreement for relocation or transition of holdings to another extranational repository in the event that the repository can no longer fulfill its mandate due to funding or shifted priorities?	No documented plan or agreement exists.	There is a formal agreement but lacking detailed plans.
	Parent Institutional Continuity Plans	What happens to the repository if the larger institution is undergoing a crisis?	There are no plans for what would happen to the repository in case of larger institutional crisis.	
	Funding Diversity	How many funding streams to the upkeep of a repository? If one or funding streams suddenly disappeared, would operations be heavily impacted?	There is one main funding stream and few or no smaller funding streams. Losing the main funding stream would shutdown the repository.	There are one or two main funding streams and a few smaller funding streams. Losing one of the main funding streams would heavily impact the repository.

Section Theme: Network

Are there peer organizations the repository can rely on in a crisis? Are there metadata and instructions on what the public should archive in case of crisis?	Professional Networks	Is the repository a member of any professional networks, national or extranational, that could participate in reciprocal data sharing agreements?	The repository is not a member of any professional networks.	The repository is a member of one or more professional networks, but there are no reciprocal data sharing agreements.
	Sibling Repositories	Is the repository a member of a larger organization with access to libraries or archives that could support dataset deposits?	There are no similar libraries, archives, or repositories in the larger organization.	There are other libraries, archives, or repositories in the larger organization but there are no explicit agreements on data rescue by them in the case of crisis.
	Public Announcements	Is the repository able to provide advanced warning to the scientific community and public about possible disruptions in service?	Notices can be posted on the website.	
	Citizen Archival	Is the repository able to facilitate citizen archivists in coordinating and archiving the repository? Can the repository find out what has been archived by citizen archivists?	There is no metadata available to the public identifying datasets that are not known to be replicated elsewhere. There are minimal lines of communication with the public.	

Section Themes Total Weighted Score

Single Repository Dataset Replication (Min=5, Max=24)	Repository Infrastructure (Min=4, Max=18)	Routine Operations (Min=3, Max=9)	Software (Min=6, Max=21)
0	0	0	0
Cybersecurity (Min=3, Max=9)	Sustainability (Min=6, Max=30)	Network (Min=4, Max=18)	
0	0	0	

isis risks to their repository. A higher score is better.

Repository Name:

Repository Parent Organization:

Date:

Project:

Scenarios

More
resiliency

Fill in this
column only!

(Yes/No question to
be asked, "No" = 0)

Class 3
User Entered
Score. (Integers
only)
Range:
1 -Class 1
2 -Class 2
3 -Class 3

Suggested
Weight
(If score > 1)

Weighted Score

Scenario Weighted
Score (Zeros are
assigned to
categories that
would be impacted
during a crisis)

1: Natural
Disaster hits in
less than 48
hours

2: Repository
defunding by
parent
organization or
primary funder,
ex. government

ts. Datasets held by another organization in the same project (ICARUS at NSF NCAR vs I

Multiple copies on multiple physical devices		1	0	Could any logical copies survive the crisis?	0	0
Three or more types of storage medium		1	0	Would any types of used storage medium survive the crisis?	0	0
Two or more offline copies		2	0	Could a secure offline copy be preserved and used after the crisis?	0	0
Three or more data facilities, geographically separated		2	0	Would a different data facility be able to maintain preserving and serving data during the crisis?	0	0
Two or more international data facilities		2	0	Would a different data facility be able to maintain preserving and serving data during the crisis?	0	0
Section Total Raw Score. (Min=5, Max=15)	0	Section Total Weighted Score. (Min=5, Max=24)	0	Section Total Weighted Scenario Score. (Min=0, Max=24)	0	0

There are multiple staff members for each role and staff in each role can cover for staff in other roles.		2	0	Are staff able to perform their duties?	0	0
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Data volume is small enough to effectively transfer all data within one day.		1	0	Is it possible to virtually move data in the time remaining?	0	0
There is capability to break down and physically move datasets in one hour notice.		1	0	Is it possible to physically move data in the time remaining?	0	0
There is redundant power/HVAC/emergency power generation for 24 hours or more.		2	0	Will the physical plant infrastructure keep cyberinfrastructure running?	0	0
Section Total Raw Score. (Min=4, Max=12)	0	Section Total Weighted Score. (Min=4, Max=18)	0	Err:539	0	0

Backups occur regularly and to one or more devices. Backups are tested on a regular schedule.		1	0	Can backups be restored from after the crisis?	0	0
SLAs are defined and publicly accessible.		1	0	Can SLAs and their deadlines be found by the public and followed during or after the crisis?	0	0
Required training for staff and time/funds for staff improvement via optional conferences, trainings, or certifications.		1	0	Can training continue after the crisis?	0	0
Section Total Raw Score. (Min=3, Max=9)	0	Section Total Weighted Score. (Min=3, Max=9)	0	Err:539	0	0

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The dataset can be made available as text (e.g. PDF file or CSV or just a plain printout or even photograph).

The software is packaged by the developer and possibly held in software repository indices. The repository has made images and backups of their current software and configuration, and can redeploy with minimal downtime.

The software is open source and has an active community supporting it.

Source code is well documented and easy to install. There are images available for deployment by non-technical users.

The software is in a common language and does not require niche software libraries and licensed compilers to execute. Code discipline is good and there is thorough test coverage of the software.

User Interface and User Experience of the software is polished for all users. Documentation is clear and thorough. Sane defaults and examples are available for the software. Programmatic access is well documented and tested.

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0

Is the primary software or alternatives available during the crisis?

0

0

1

0

Is the technical expertise to package the software systems the repository runs on available?

0

0

2

0

Is the technical expertise creating the software still available?

0

0

1

0

Can the software be installed by someone with no experience in an acceptable timeline?

0

0

1

0

Can the software be maintained by someone with no experience in an acceptable time frame?

0

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1

0

Can the software be used by someone with no experience in an acceptable time frame?

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0

Section Total Raw Score. (Min=6, Max=18)	0	Section Total Weighted Score. (Min=6, Max=21)	0	Section Total Weighted Scenario Score. (Min=1, Max=21)	0	0
There is real time logging with alerts at the operating systems level and repository systems level. Alerts will notify staff of anomalous changes such as increased deletions or access. Logging is available for all facilities and are monitored before an update of off-site copies. There are some accounts with higher permissions for modifying datasets and systems. There is multi factor authentication required for authentication and executing critical tasks like deletion of data. Logs around stopping of systems and deletion of deposits are reviewed periodically. There are cybersecurity professionals available through a larger organization and a cybersecurity professional on staff.		1	0	Is logging still working and being monitored during the crisis to ensure normal operations are occurring?	0	0
		1	0	Are permissions being maintained and authorized access and actions being controlled to ensure normal operations during the crisis?	0	0
		1	0	Would cybersecurity staff be able to mitigate issues of the crisis?	0	0
		Section Total Weighted Score. (Min=3, Max=9)	0	Section Total Weighted Scenario Score. (Min=0, Max=9)	0	0

Datasets/data systems are regularly modernized with links to tools for accessing them. There is input from the community on how modernization should be done. There are plans for a systematic shutdown of systems, transfer of assets, and care for staff. There is a signed formal agreement with another data repository, with documented plans as to how to execute the transition. There is a signed formal agreement with another data repository, with documented plans as to how to execute the transition. There is a clear plan of what would happen to the repository in case of larger institutional crisis, including merging with another repository or moving to another unit in the institution. There are multiple, roughly equivalent funding streams such that losing one or two is not a critical blow to operations.						
	2	0	Can data modernization or migration happen after the crisis?	0	0	
	1	0	Can repository shutdown plans be enacted during the scenario?	0	0	
	2	0	Have the continuity agreement and plans been executed before or possibly during the scenario?	0	0	
	2	0	Have the continuity agreement and plans been executed before or possibly during the scenario?	0	0	
	1	0	Can the parent institution execute continuity plans during the crisis?	0	0	
	2	0	Will funding streams continue during and after the crisis?	0	0	
Section Total Raw Score. (Min=6, Max=18)	0	Section Total Weighted Score. (Min=6, Max=30)	0	Err:539	0	0

Rubric Total Weighted Score < 43	43 <= Rubric Total Weighted Score <= 86	86 < Rubric Total Weighted Score
Low resiliency	Moderate resiliency	High resiliency

Err:539

Err:539

3: Current cyberattack, cybercrime, or other loss of control over cyberinfrastructure

4: Loss of technical expertise critical to running repository

UC Davis) must fill this out

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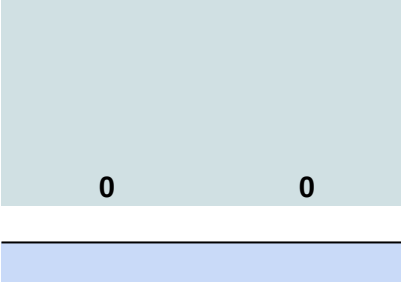
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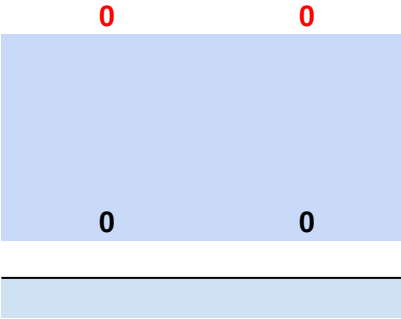


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um possible scores for scenarios are:

Err:539

Err:539

The Data Impact and Recovery Scorecard (DIRS) is used to determine metadata resiliency ;
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Repository Crisis Scorecards Instructions

Repository Descriptors			Resiliency Metric	
			No resiliency	
Big Picture Question	Descriptor	Descriptor definition	Class 1	Class 2
Section Theme: Data and Metadata				
<u>Can the data be reconstructed if the central servers are offline? Look at https://metadata-hub.org/</u>				
	Data Management Plans	Is there a data management plan (DMP) attached to the dataset that describes all related publications and datasets?	There is no DMP attached to the dataset.	There is a incomplete or inaccurate DMP attached to the dataset. It doesn't address all the places where the dataset is described.
	PID Coverage and Landing Pages	Does each resource have a persistent identifier (PID)? Does each resource have its own landing page?	No PIDs used in this repository; no landing pages or sitemap.	There are either PIDs and landing pages or sitemaps but not both.
	Metadata Completeness	How complete are the metadata currently? How complete do they need to be for someone to reuse/reconstruct the data/metadata? e.g., OAIS reference model completeness	Every resource has incomplete metadata for required and optional fields.	All resources have complete metadata for required fields, some have metadata for optional fields.
	FAIR Compliance	In particular, does the (meta)data follow recognised standards and are all semantic resources used to describe the datasets such as vocabularies, metadata schemas, ontologies, etc.?	FAIR is not considered.	Dataset considers FAIR, but is only human readable.
	CARE Compliance or Ethical Usage Considerations	Are ethical considerations taken regarding a dataset, including ownership and governance of the dataset by and for the people the data comes from?	Datasets could have direct impact on people but have not been considered through an ethical framework, such as CARE principles.	Dataset has considered ethical frameworks and indigenous community standards or other ethical considerations implemented, such as Local Context notices.

Would it be straightforward for others to understand and reuse your data and metadata?

Data Deprecation Plans

Does the repository have a process for evaluating and deprecating datasets if needed?

Datasets are never evaluated for deprecation.

Datasets are deprecated ad hoc upon request from depositors. No formal process exists for evaluating dataset deprecation.

Section Theme: Impact

Third Party Application Support Through Transition

Can third party applications still access data needed if access is lost e.g., earthquake detections, tsunami detections, Marine Protected Area monitoring, data pushed to the WMO for forecast models, etc?

There are no alternate paths for datasets to be delivered and/or transformed to users.

There are tools or alternate paths to the data but they may not be free and may not be public.

Audience Impact

Are warnings still able to be issued to the public by others based on datasets and analysis held by the repository, e.g warnings to floods, hurricanes, tornadoes, air quality, etc?

There are no other agencies able to perform similar actions, i.e. issue similar warnings.

There are agencies that provide a paid service for similar actions, i.e. pay for warnings.

Federated Networks

Will the loss of access cause holes in networks relying on data hosted by members?

The repository is the only host that provides access for the network, and loss of access causes a hole.

There are other organizations in the network that mirror some but not all of the repository's holdings to provide a partial solution.

How would the loss of repository

Replicating Other Repositories

If the repository is replicating datasets for other repositories such that if the other repository(s) were to go offline, would there be no access to the datasets?

The repository is hosting the only known replica of another repository, and the loss of both repositories would lead to the complete loss of access to the deposits.

The repository is one of many hosting replicas of another repository, such that the loss of this repository and the replicated repository causes a partial loss of access to the deposits.

and societal impacts. A higher score is better/more imp Repository Name:

Repository Parent Organization:

Date:

Project:

**More
resiliency**

*Fill in this
column only!*

User Entered
Score. (Integers
only) Score

Class 3

Range:
0 -N/A (Impact only)
1 -

Suggested
Weight
(If score > 1)

Weighted Score

Class 1

2 -Class 2

3 -Class 3

health-ui.vercel.app/ for a summary of repository metadata

There is a DMP attached to the dataset, and it describes other related publications and datasets.	1	0
Every resource has a PID and landing page, and the repository has a sitemap including all resources.	2	0
All resources have complete metadata for required fields and optional fields.	2	0
Dataset is as FAIR compliant as possible including machine actionable.	1	0
Dataset has ethical frameworks implemented and metadata allows for fine grained data governance, and for ethical metadata to persist with the datasets; or dataset has considered ethical frameworks and they are not applicable.	1	0

Datasets are periodically evaluated for deprecation according to a published schedule, and feedback is solicited from the community as to appropriateness.		1	0
Section Total Raw Score. (Min=6, Max=18)	0	Section Total Weighted Score. (Min=6, Max=24)	0

There are free public tools and alternate pipelines for delivering datasets that can be simply reconfigured to use a different endpoint.		2	0
There are multiple other agencies, including international agencies, that can perform similar actions for free with similar reach.		2	0
There are one or more organizations in the network that hold a full replica of the repository that is regularly synced for changes.		1	0
The repository is one of many hosting replicas of another repository, such that the loss of this repository and the replicated repository does not cause any loss of access to the deposits.		1	0

Section Total Raw Score. (Min=4, Max=12)	0	Section Total Weighted Score. (Min=4, Max=18)	0
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Repository Resilience Scorecard (RRS)					RCS, Version 1.02 Released 2025-04-15									
RRS Raw and Weighted Score					RRS Section x Scenario Weighted Min									
	Raw Min	Raw Max	Weighted Max	Thirds	RRS Total Scores					Scenario 1	Scenario 2	Scenario 3	Scenario 4	
Section 1	5	15	24	6.33	Raw Min	31				Section 1	3	0	5	5
Section 2	4	12	18	4.67	Raw Max	93				Section 2	Err:539	3	Err:539	3
Section 3	3	9	9	2.00	Weighted Ma	129				Section 3	2	0	Err:539	Err:539
Section 4	6	18	21	5.00	Lower Third	43				Section 4	6	6	6	1
Section 5	3	9	9	2.00	Upper Third	86				Section 5	2	3	1	0
Section 6	6	18	30	8.00	Lower Third v	64				Section 6	6	4	Err:539	Err:539
Section 7	4	12	18	4.67	Upper Third v	97				Section 7	4	Err:539	2	Err:539
Total	31	93	129							Total	Err:539	Err:539	Err:539	Err:539
RRS Scenario Combinatorics Score					RRS Section x Scenario Weighted Max									
	Weighted Min	Weighted Max	Thirds	Sum						Scenario 1	Scenario 2	Scenario 3	Scenario 4	
1&2: Nat Disaster & Defundin	Err:539	Err:539	Err:539	Err:539						Section 1	18	6	24	24
1&3: Nat Disaster & Cyberatt	Err:539	Err:539	Err:539	Err:539						Section 2	15	12	15	12
1&4: Nat Disaster & Tech Ex	Err:539	Err:539	Err:539	0						Section 3	6	0	6	6
2&3: Defunding & Cyberattac	Err:539	Err:539	Err:539	0						Section 4	21	21	21	3
2&4: Defunding & Tech Exp	Err:539	Err:539	Err:539	Err:539						Section 5	6	9	3	0
3&4: Cyberattack & Tech Exp	Err:539	Err:539	Err:539	Err:539						Section 6	30	18	24	18
1&2&3: Nat Disaster & Defun	Err:539	Err:539	Err:539	Err:539						Section 7	18	15	9	15
1&2&4: Nat Disaster & Defun	Err:539	Err:539	Err:539	Err:539						Total	114	81	102	78
1&3&4: Nat Disaster & Cyber	Err:539	Err:539	Err:539	Err:539										
2&3&4: Defunding & Cyberat	Err:539	Err:539	Err:539	Err:539										
1&2&3&4: Nat Disaster & De	Err:539	Err:539	Err:539	Err:539										
Data Impact and Recovery Scorecard (DIRS)					DIRS Total Scores									
DIRS Raw and Weighted Score					DIRS Total Scores									
	Raw Min	Raw Max	Weighted Max	Thirds	Raw Min	10								
Section 1	6	18	24	6.00	Raw Max	30								
Section 2	4	12	18	4.67	Weighted Ma	42								
Scenario Combinatorics Score											Section Themes Total Weighted Score			
Err:539	Err:539	Err:539	Err:539	Err:539	Err:539	Err:539	Err:539	Err:539	Err:539	Err:539	Data and Metadata (Min=6, Max=24)	Impact (Min=4, Max=18)		
											0	0		